MATHEMATICS AND STATISTICS

Time: 3 Hrs. **Max. Marks: 80**

General Instructions:

(1) All questions are compulsory.
(2) There are 6 questions divided into two sections.
(3) Write answers of Section-I and Section-II in the same answer book.
(4) Use of logarithmic tables is allowed. Use of calculator is not allowed.
(5) For L.P.P. graph paper is not necessary. Only rough sketch of graph is expected.
(6) Start answer to each question on a new page.
(7) For each multiple choice type of question, it is mandatory to write the correct answer along with its alphabetical letter eg. (a) ……../(b)……../(c)……../(d)……. No mark(s) shall be given if "ONLY" the correct answer or the alphabet of the correct answer is written. Only the first attempt will be considered for evaluation.

SECTION – I

Q.1. (A) Select and write the correct answer of the following multiple choice type of questions (1 mark each): **(6)[12]**

(i) if p: He is intelligent
q: He is strong.
are statements in logic. Then symbolic form of statement.
'It is false that, he is intelligent or strong' is _____.

(a) $\sim p \vee \sim q$ (b) $\sim (p \wedge q)$ (c) $\sim (p \vee q)$ (d) $p \vee \sim q$

(ii) If $y = e^{\log x}$ then $\frac{dy}{dx}$ = _____.

(a) $\frac{e^{\log x}}{x}$ (b) $\frac{1}{x}$ (c) 0 (d) $\frac{1}{2}$

(iii) If $x = 2at^2$, $y = 4at$ then $\frac{dy}{dx}$ = _____.

(a) $\frac{1}{t}$ (b) $\frac{-1}{2at^2}$ (c) $\frac{1}{2at^3}$ (d) $\frac{1}{4at^3}$

(iv) If elasticity of demand $\eta = 1$ then demand is _____.

(a) constant (b) inelastic (c) unitary elastic (d) elastic

(v) The statement $p \rightarrow q$ is equivalent to_____.

(a) $p \rightarrow \sim q$ (b) $\sim p \vee q$ (c) $\sim p \rightarrow \sim q$ (d) $p \vee \sim q$

(vi) The matrix $\begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}$ is _____.

(a) identity matrix (b) scalar matrix
(c) null matrix (d) column matrix

(B) State whether the following statements are true or false (1 mark each): **(3)**

(i) The area of the portion lying above the X-axis is positive.

(ii) The order of highest derivative occurring in the differential equation is called degree of the differential equation.

(iii) If A and B are conformable for the product AB then $(AB)^T = A^T B^T$.

(C) Fill in the following blanks (1 mark each): **(3)**

(i) If $y = x \cdot \log x$ then $\frac{dy}{dx} =$ _____.

(ii) If $\int_0^a 3x^2 dx = 8$, then a = _____.

(iii) Order and degree of differential equation are always _____ integers.

Q.2. (A) Attempt any TWO of the following (3 marks each): **(6)[14]**

(i) Write the converse, inverse and contrapositive of the statement, "If he studies, then he will go to college".

(ii) Find x and y if $\left\{4\begin{bmatrix}2 & -1 & 3\\ 1 & 0 & 2\end{bmatrix} - \begin{bmatrix}3 & -3 & 4\\ 2 & 1 & 1\end{bmatrix}\right\}\begin{bmatrix}2\\ -1\\ 1\end{bmatrix} = \begin{bmatrix}x\\ y\end{bmatrix}$.

(iii) Find the area of the region bounded by the curve $y = x^2 + 1$, lines $x = 0$, $x = 3$ and X-axis.

(B) Attempt any TWO of the following (4 marks each): **(8)**

(i) If $x^3 y^3 = x^2 - y^2$, then find $\frac{dy}{dx}$.

(ii) Find MPC, MPS, APC and APS, if the expenditure E_e of a person with income I is given as $E_e = (0.0003)\, I^2 + (0.075)\, I$ when $I = 1000$

(iii) Find the population of a city at any time 't' given that the rate of increase of population is proportional to the population at that instant and that in a period of 40 years the population increased from 30,000 to 40,000.

Q.3. (A) Attempt any TWO of the following questions (3 marks each): **(6)[14]**

(i) Using the truth table, examine whether the given statement pattern is tautology, contradiction or contingency.

$p \rightarrow (q \vee r)$

(ii) Find the inverse of matrix $\begin{bmatrix}3 & -1\\ 2 & -1\end{bmatrix}$ by adjoint method.

(iii) Evaluate: $\int \frac{1}{x^2 + 4x - 5}\, dx$

(B) Attempt any ONE of the following (4 marks each): **(4)**

(i) Evaluate: $\int x^3 e^{x^2}\, dx$

(ii) Evaluate : $\int_2^7 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{9 - x}}\, dx$

(C) Attempt any ONE of the following questions (Activity): **(4)**

(i) A metal wire of 36 cm long is bent to form a rectangle. Find its dimensions when its area is maximum by completing following activity:

Solution:

Let x and y be the length breadth of rectangle, then

$x + y =$ ☐

then area of rectangle = $18x - x^2$

Let $f(x)$ be the area of rectangle

$f(x) = 18x - x^2$

$f'(x) =$ ☐

$f''(x) = -2$

For extreme value $f'(x) = 0$

$x =$ ☐

Now $f'(q) < 0$

$\therefore$ $f(x)$ [Area] is maximum when dimensions (length and breadth are) ☐

(ii) Solve the following differential equation $x^2 y\,dx - (x^3 + y^3)\,dy = 0$ by filling the boxes in the following activity:

Solution:

$x^2 y dx - (x^3 + y^3)dy = 0$

This is homogeneous equation

$\therefore$ we put $y = vx$

$\therefore$ $\frac{dy}{dx} =$ ☐

From given equation

$$\frac{dy}{dx} = \frac{x^2 y}{x^3 + y^3}$$

$$v + x \cdot \frac{dv}{dx} = \frac{x^2 \cdot xv}{x^3 + v^3 x^3}$$

$$= \frac{v}{1 + v^3}$$

$$x \cdot \frac{dv}{dx} = \frac{v}{v^3 + 1} - v$$

$$= \frac{v - v^4 - v}{v^3 + 1}$$

$\therefore$ $$x\frac{dv}{dx} = \frac{-v^4}{v^3 + 1}$$

$\therefore$ ☐ $dv = \frac{-dx}{x}$

Integrating both sides

$$\int\left(\frac{1}{v} + \frac{1}{v^4}\right)dv = -\int\frac{dx}{x}$$

☐ + log c = –log x

$$\log cvx = \frac{1}{3v^3}$$

Substituting the value of v we get

☐ is the general solution.

SECTION – II

Q.4. (A) Select and write the correct answer of the following multiple choice type of questions (1 mark each): **(6)[12]**

(i) An agent who gives guarantee to his principal that the party will pay the sale price of goods is called _____.

(a) Auctioneer (b) Delcredere agent (c) Factor (d) Broker

(ii) X is a number obtained on upper most face when a fair die is thrown then E(x) = _____.

(a) 3 (b) 3.5 (c) 4 (d) 4.5

(iii) The cost of living Index number using Aggregate Expenditure method is given by _____.

(a) $\frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$ (b) $\frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$

(c) $\frac{\sum p_1 q_1}{\sum p_1 q_0} \times 100$ (d) $\frac{\sum p_0 q_0}{\sum p_1 q_0} \times 100$

(iv) In sequencing an optimal path minimizes _____.

(a) Elapsed time (b) Idle time

(c) Both (a) and (b) (d) Ready time

(v) The marked price is also called as _____.

(a) Cost price (b) Selling price (c) List price (d) Invoice price

(vi) If $X \sim B\left(20, \frac{1}{10}\right)$, then $E(x)$ = _____.

(a) 2 (b) 5 (c) 4 (d) 3

(B) State whether the following statements are true or false (1 mark each): **(3)**

(i) In an assignment problem, if number of columns is greater than number of rows then a dummy column is added.

(ii) The optimum value of the objective function of LPP occurs at the centre of the feasible solution.

(iii) Regression equation of X on Y is $y - \bar{y} = b_{yx}\left(x - \bar{x}\right)$.

(C) Fill in the blanks (1 mark each): **(3)**

(i) If X is discrete random variable which takes the values $x_1, x_2, \ldots, x_n$ then $\sum_{i=1}^{n} p(x_i)$ = _____.

(ii) The constraint that a factory has to employ more women (y) than men (x) is given by _____.

(iii) If $F(x)$ is distribution function of discrete r.v. X with p.m.f.

$p(x) = \frac{x-1}{3}$, for $x = 1, 2, 3$

$= 0$, otherwise

then $F(4)$ = _____.

Q.5. (A) Attempt any TWO of the following questions (3 marks each): **(6)[14]**

(i) Using 1995 as base year in the following problem; find the price index number by aggregate method.

Commodity	A	B	C	D	E
Price (in ₹) in 1995	42	30	54	70	120
Price (in ₹) in 2005	60	55	74	110	140

(ii) Four new machines M_1, M_2, M_3 and M_4 are to be installed in a machine shop. There are five vacant places A, B, C, D and E available. Because of limited space, machine M_2 cannot be placed at C and M_3 cannot be placed at A. The cost matrix is given below:

Machines	Places				
	A	B	C	D	E
M_1	4	6	10	5	6
M_2	7	4	–	5	4
M_3	–	6	9	6	2
M_4	9	3	7	2	3

Find the optimal assignment schedule.

(iii) The following table shows the production of gasoline in U.S.A. for the years 1962 to 1976. Obtain trend values using 5-yearly moving averages:

Year	Production (Million Barrels)	Year	Production (Million Barrels)
1962	0	1970	6
1963	0	1971	7
1964	1	1972	8
1965	1	1973	9
1966	2	1974	8
1967	3	1975	9
1968	4	1976	10
1969	5		

(B) Attempt any TWO of the following questions (4 marks each): **(8)**

(i) There are 10% defective items in a large bulk of items. What is the probability that a sample of 4 items will include not more than one defective item?

(ii) Minimize $z = 4x + 2y$
Subject to $3x + y \geq 27$
$x + y \geq 21$
$x \geq 0, y \geq 0$
Solve above LPP by graphical method.

(iii) A building is insured for 75% of its value. The annual premium at 0.70 percent amounts to ₹ 2,625. If the building is damaged to the extent of 60% due to fire, how much can be claimed under policy?

Q.6. (A) Attempt any TWO of the following questions (3 marks each): **(6)[14]**

(i) If $\sum p_0q_0 = 140$, $\sum p_0q_1 = 200$, $\sum p_1q_0 = 350$, $\sum p_1q_1 = 460$, find Laspeyre's price index number, Pasche's price index number and Dorbish-Bowley's price Index number.

(ii) For a bivariate data $\bar{x} = 53$, $\bar{y} = 28$, $b_{YX} = -1.2$; $b_{XY} = -0.3$, find
(a) Correlation coefficient between X and Y.
(b) Estimate Y for X = 50

(iii) Defects on plywood sheet occur at random with the average of one defect per 50 sq.ft. Find the probability that such a sheet has (i) no defect (ii) at least one defect
[use $e^{-1} = 0.3678$]

(B) Attempt any ONE of the following questions (4 marks each): **(4)**

(i) Find the sequence that minimizes the total elapsed time to complete the following jobs. Each job is processed in order AB.

	Jobs (Processing time in minutes)						
	I	II	III	IV	V	VI	VII
Machine A	12	6	5	11	5	7	6
Machine B	7	8	9	4	7	8	3

Find the total elapsed time and the idle times for both the machines.

(ii) From the following data, find the regression equation of Y on X and estimate Y when X = 10

X	1	2	3	4	5	6
Y	2	4	7	6	5	6

(C) Attempt any ONE of the following questions (Activity): **(4)**

(i) A bill was drawn on 14th April for ₹ 7,000 and was discounted on 6th July at 5% p.a. The banker paid ₹ 6,930 for the bill. Find the period of the bill by filling boxes in the following activity.

Solution:

SD = ₹ 7,000, r = 5%, CV = 6,930

BD = ☐

$BD = \frac{SD \times n \times r}{100}$

n = ☐ days

Number of days to be counted from date of discounting i.e., 6th July

July	Aug.	Sep.	Total

∴ Period of Bill is ☐ month

(ii) Find the equation of trend line for the following data showing production of steel (in millions of tonnes) for years 1976 to 1986 by filling boxes in the following activity:

Year	1976	1977	1978	1979	1980	
Production	0	4	4	2	6	
Year	1981	1982	1983	1984	1985	1986
Production	8	5	9	4	10	10

Solution:

Year (t)	**Production y_t**	**u = t – 1981**	**u^2**	**uy_t**
1976	0	–5	25	0
1977	4	–4	16	–16
1978	4	–3	9	–12
1979	2	–2	4	–4
1980	6	–1	1	–6
1981	8	0	0	0
1982	5	1	1	5
1983	9	2	4	18
1984	4	3	9	12
1985	10	4	16	40
1986	10	5	25	50
Total	**$\sum y_t = 62$**	**$\sum u = 0$**	**$\sum u^2 = 110$**	**$\sum uy_t = 87$**

u = t – 1981

Let equation of trend line be

y_t = ☐ ….(1)

$\sum y_t = na' + b'\sum u$ ….(2)

$\sum uy_t = a'\sum u + b'\sum u^2$ ….(3)

a′ = ☐

b′ = ☐

∴ Equation of trend line is y_t = ☐